

USER GUIDE

Ladder on Box Solver for HP15c CE software pac

(rev1)

Pepin Torres, P.E.

pepin(dot)torres{at}gmail(dot)com

Introduction:

This software pac solves the classic problem of a ladder resting on a square box¹. This program will solve the problem in two ways: (1) Plugging and chugging and (2) using the HP15c SOLVE function.

Who is this for?

This is for anyone who is ever in dire need of calculating the distance between the tip of the ladder down to the top of the box on which the ladder is resting.

In all honesty, this is just a trivial program for one very particular type of problem – unless you really need to find out how high you can climb on a ladder of length L resting on a square box of height c . However, it does two things for me: helps me get experience coding with the SOLVE function and gives me some exposure on how to use it to find the roots of a function. There is also a nostalgia aspect which I discuss at the end of this document.

Excel file includes listing, comments, stack states, and sanity check examples.

Note: The program is short enough that inputting by hand is not objectionable plus the added benefit of not having to blow away existing programs. The .mem file is provided for those who would rather flash it.

To calculate height (plug and chug):

Prerequisites:

Stack:

y-register: Length L of ladder

x-register: height c of square box

Registers: None

Usage:

Press **GSB A**

Output:

x-register: Wall distance from top of box to top of ladder

To find floor distance from bottom of ladder to the bottom of box press $1/x$

Registers used: R0, R1, R2, R3, R4

¹ See this [great page](#) and [video](#) by Presh Tawalkar for the definition of the puzzle and its solution.

To calculate height (using SOLVE):

Prerequisites:

Stack:

y-register: Length L of ladder

x-register: height c of square box

Registers: None

Usage:

Press **GSB B**

Output:

x-register: Wall distance from top of box to top of ladder

To find floor distance from bottom of ladder to the bottom of box press $1/x$

Registers used: R0, R1, R2, R3, R4

Why tho?

In the summer of 1989 as a 10th grader, I attended a science and engineering summer camp at a university. Even when the classes very crazy advanced, it was one of the greatest summers of my life. It defined me as a person and gave me the confidence to get into the sciences. On the last day, the math professor presented us with the ladder-on-box problem and just said: go solve it. We spent a long time trying to figure it out as a group to no avail by the time the bell rang. That problem stayed (unsolved) in the back of my head all these years. A couple of years ago, while chatting about the problem with a friend from that summer camp, I decided to finally solve it. So, one day I sat down and didn't get up until I had it figured out. So, 35-odd years later I solved *the last unknown* of that summer camp. Now that I am obsessed with writing applications for the HP15c CE, I decided to include this mathematical curiosity in the HP15c CE "engineering basics" software pac I have been working on.

Closing note: THIS PROGRAM WORKS AS EXPECTED ON THE HP15C CALCULATOR. HOWEVER, SOME OF THE SIMULATORS TEND TO FIND THE ROOTS IN THE REVERSE ORDER, OR YIELD A COMPLETELY ERRONEOUS SOLUTION.